



UNIVERSIDAD NACIONAL DE INGENIERÍA

Facultad de Ingeniería Eléctrica y Electrónica

Cálculo Integral - BMA02P

Tarea N°01

Solucionario PC1

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Lima, Perú 2026 - I



1. $\int \frac{\sqrt{x^2 + 1} (\ln(x^2 + 1) - 2\ln(x))}{x^4} dx$

Solución.

Dándole la forma a la integral

$$I = \int \frac{\sqrt{x^2 + 1} \left(\ln \left(\frac{x^2 + 1}{x^2} \right) \right)}{x^4} dx$$

Se realiza el siguiente cambio de variable

$$z = \frac{x^2 + 1}{x^2} \rightarrow dz = -\frac{2}{x^3} dx$$

Sustituyendo en la integral

$$I = -\frac{1}{2} \int \sqrt{\frac{x^2 + 1}{x^2}} \cdot \ln(z) dz \rightarrow I = -\frac{1}{2} \int z^{1/2} \cdot \ln(z) dz$$

Por integración por partes

$$u = \ln(z) \rightarrow du = \frac{dz}{z}$$
$$dv = z^{1/2} dz \rightarrow v = \frac{2}{3} z^{3/2}$$

Resolviendo la integral

$$I = -\frac{1}{2} \left[\frac{2}{3} z^{3/2} \ln(z) - \frac{2}{3} \int \frac{z^{3/2}}{z} dz \right] \rightarrow I = -\frac{1}{2} \left[\frac{2}{3} z^{3/2} \ln(z) - \frac{4}{9} z^{3/2} + C \right]$$

Devolviendo a su variable original

$$I = -\frac{1}{3} \left[\frac{x^2 + 1}{x^2} \right]^{3/2} \cdot \ln \left(\frac{x^2 + 1}{x^2} \right) + \frac{2}{9} \left[\frac{x^2 + 1}{x^2} \right]^{3/2} + C$$



$$2. \int \frac{e^{\ln(\cos(2x)) - \ln(\sec(2x))} + e^{\cos(2x) + 4\ln(x^{1/4} \sin(2x))}}{e^{(\cos(2x) - 3\ln(\csc(2x)))}} dx$$

Solución.

Reduciendo por propiedad de logaritmos

$$I = \int \frac{e^{\ln(\cos(2x))} \cdot e^{\ln(\cos(2x))} + e^{\cos(2x)} \cdot e^{4\ln(x^{1/4} \sin(2x))}}{e^{\cos(2x)} \cdot e^{3\ln(\sin(2x))}} dx$$

$$I = \int \frac{\cos^2(2x) + e^{\cos(2x)} [x^{1/4} \sin(2x)]^4}{e^{\cos(2x)} \cdot \sin^3(2x)} dx$$

Separando en dos integrales

$$I = \int \frac{\cos^2(2x)}{e^{\cos(2x)} \sin^3(2x)} dx + \int \frac{x \sin^4(2x)}{\sin^3(2x)} dx$$

$$I = \int \frac{\cos^2(2x)}{e^{\cos(2x)} \sin^3(2x)} dx + \int x \cdot \sin(2x) dx$$

Resolviendo por IPP el segundo término entonces

$$I = \int \frac{\cos^2(2x)}{e^{\cos(2x)} \sin^3(2x)} dx - \frac{x}{2} \cos(2x) + \frac{1}{4} \sin(2x) + C$$

$$3. \int \frac{x \cosh^2(x) \sinh(x) - \cosh^3(x) + x^2 \sinh(2x) - 2x \cosh^2(x)}{x(\cosh^3(x) - x^3)} dx$$

Solución.

Sea el cambio de variable

$$u = \frac{\cosh x}{x} \rightarrow du = \frac{x \sinh x - \cosh x}{x^2} dx$$

Sustituyendo en la integral

$$I = \int \frac{x(ux)^2 \sinh x - (ux)^3 + 2x^3 u \sinh x - 2x(ux)^2}{x((ux)^3 - x^3)} \cdot \frac{x^2}{x \sinh x - \cosh x} du$$

$$I = \int \frac{x^3 u^2 \sinh x - u^3 x^3 + 2x^3 u \sinh x - 2u^2 x^3 x}{x^4(u^3 - 1)} \cdot \frac{x^2}{x \sinh x - \cosh x} du$$

Agrupando el x^3 en el numerador

$$I = \int \frac{((u^2 + 2u) \sinh x - u^3 - 2u^2)}{(u^3 - 1)} \cdot \frac{du}{\sinh x - u}$$

Factorizando

$$I = \int \frac{(u^2 + 2u)(\sinh x - u)}{u^3 - 1} \cdot \frac{du}{(\sinh x - u)} \rightarrow I = \int \frac{u^2 + 2u}{u^3 - 1} du$$

Por fracciones parciales

$$\frac{u^3 + 2u}{(u - 1)(u^2 + u + 1)} = \frac{A}{u - 1} + \frac{Bu + C}{u^2 + u + 1}$$

Resolviendo tenemos que

$$A = C = 1 \quad \wedge \quad B = 0$$

Entonces

$$I = \int \frac{du}{u - 1} + \int \frac{du}{u^2 + u + 1}$$

$$I_1 = \int \frac{du}{u - 1} = \ln|u - 1| + C_1$$

$$I_2 = \int \frac{du}{u^2 + u + 1} = \frac{2}{\sqrt{3}} \arctan\left(\frac{2u + 1}{\sqrt{3}}\right) + C_2$$

Reemplazando en la integral y devolviendo a su variable original

$$I = \ln \left| \frac{\cosh x}{x} - 1 \right| + \frac{2}{\sqrt{3}} \arctan \left(\frac{\frac{2 \cosh x}{x} + 1}{\sqrt{3}} \right) + C$$

4. $\int \frac{e^{\cos(x)} \cdot (x \operatorname{sen}^3(x) + \cos(x))}{\cos^2(x)} dx$

Solución.

Separando en dos integrales

$$I = \int x e^{\cos(x)} \frac{\operatorname{sen}^3(x)}{\cos^2(x)} dx + \int \frac{e^{\cos(x)}}{\cos(x)} dx$$

$$I_1 = \int x e^{\cos(x)} \frac{\operatorname{sen}(x) \cdot \operatorname{sen}^2(x)}{\cos^2(x)} dx \rightarrow I = \int x e^{\cos(x)} \frac{\operatorname{sen}(x) \cdot (1 - \cos^2(x))}{\cos^2(x)} dx$$

$$I_1 = \int x e^{\cos(x)} \operatorname{sen}(x) \sec^2(x) dx - \int x e^{\cos(x)} \operatorname{sen}(x) dx$$

Aplicando IPP al segundo término de I_1 entonces

$$I = \int x e^{\cos(x)} \operatorname{sen}(x) \sec^2(x) dx + x e^{\cos(x)} - \int e^{\cos(x)} dx + \int \frac{e^{\cos(x)}}{\cos(x)} dx + C$$

5. $\int \frac{e^{32x}}{(e^{8x} + 1)^3} dx$

Solución.

Sea el cambio de variable

$$u = e^{8x} + 1 \rightarrow du = 8e^{8x} dx$$

Sustituyendo en la integral

$$I = \int \frac{(u-1)^4}{u^3} \cdot \frac{du}{8(u-1)} \rightarrow I = \frac{1}{8} \int \frac{(u-1)^3}{u^3} du$$

Expandiendo y resolviendo

$$I = \frac{1}{8} \left[u - 3 \ln|u| - \frac{3}{u} + \frac{1}{2u^2} \right] + C$$

Devolviendo a su variable original

$$I = \frac{1}{8} \left[e^{8x} + 1 - 3 \ln|e^{8x} + 1| - \frac{3}{e^{8x} + 1} + \frac{1}{2(e^{8x} + 1)^2} \right] + C$$

6.
$$\int \frac{\operatorname{sen}^3(x)\cos(x) + \operatorname{sen}(2x) - \cos(x)}{(\operatorname{sen}^2(x) + 4\operatorname{sen}(x) + 5)^4} dx$$

Solución.

Sea el cambio de variable

$$u = \operatorname{sen}(x) \rightarrow du = \cos(x)dx$$

Sustituyendo en la integral

$$I = \int \frac{u^3 + 2u - 1}{(u^2 + 4u + 5)^4} du \rightarrow I = \int \frac{u^3 + 2u - 1}{((u + 2)^2 + 1)^4} du$$

Siguiente cambio de variable

$$v = u + 2 \rightarrow dv = du$$

Reemplazando en la integral

$$I = \int \frac{(v - 2)^3 + 2(v - 2) - 1}{(v^2 + 1)^4} dv \rightarrow I = \int \frac{v^3 - 6v^2 + 14v - 13}{(v^2 + 1)^4} dv$$

Último cambio de variable

$$\tan(\phi) = v \rightarrow \sec^2(\phi)d\phi = dv$$

Sustituyendo en la integral

$$I = \int \frac{\tan^3(\phi) - 6\tan^2(\phi) + 14\tan(\phi) - 13}{\sec^8(\phi)} d\phi$$

$$I = \int [\operatorname{sen}^3(\phi)\cos^3(\phi) - 6\operatorname{sen}^2(\phi)\cos^4(\phi) + 14\operatorname{sen}(\phi)\cos^5(\phi) - 13\cos^6(\phi)] d\phi$$

$$I_1 = \int \operatorname{sen}^3(\phi)\cos^3(\phi)d\phi = -\frac{\cos^4(\phi)}{4} + \frac{\cos^6(\phi)}{6} + C_1$$

$$I_2 = \int \operatorname{sen}^2(\phi)\cos^4(\phi)d\phi = \frac{1}{16}\phi + \frac{1}{64}\operatorname{sen}(2\phi) - \frac{1}{64}\operatorname{sen}(4\phi) - \frac{1}{192}\operatorname{sen}(6\phi) + C_2$$

$$I_3 = \int \operatorname{sen}(\phi)\cos^5(\phi)d\phi = -\frac{\cos^6(\phi)}{6} + C_3$$

$$I_4 = \int \cos^6(\phi)d\phi = \frac{5}{16}\phi + \frac{15}{64}\operatorname{sen}(2\phi) + \frac{3}{64}\operatorname{sen}(4\phi) + \frac{1}{192}\operatorname{sen}(6\phi) + C_4$$

Colocando en la integral

$$I = -\frac{\cos^4(\phi)}{4} + \frac{\cos^6(\phi)}{6} - 6 \left[\frac{1}{16}\phi + \frac{1}{64}\operatorname{sen}(2\phi) - \frac{1}{64}\operatorname{sen}(4\phi) - \frac{1}{192}\operatorname{sen}(6\phi) \right] \\ + 14 \left[-\frac{\cos^6(\phi)}{6} \right] - 13 \left[\frac{5}{16}\phi + \frac{15}{64}\operatorname{sen}(2\phi) + \frac{3}{64}\operatorname{sen}(4\phi) + \frac{1}{192}\operatorname{sen}(6\phi) \right]$$

$$I = -\frac{71}{16}\phi - \frac{\cos^4\phi}{4} - \frac{13}{6}\cos^6\phi - \frac{201}{64}\operatorname{sen}(2\phi) - \frac{33}{64}\operatorname{sen}(4\phi) - \frac{7}{192}\operatorname{sen}(6\phi)$$



Devolviendo a su variable original

$$\begin{aligned} I = & -\frac{71}{16} \arctan(\operatorname{sen}(x) + 2) - \frac{\cos^4(\arctan(\operatorname{sen}(x) + 2))}{4} - \frac{13}{6} \cos^6(\arctan(\operatorname{sen}(x) + 2)) \\ & - \frac{201}{64} \operatorname{sen}(2 \arctan(\operatorname{sen}(x) + 2)) - \frac{33}{64} \operatorname{sen}(4 \arctan(\operatorname{sen}(x) + 2)) \\ & - \frac{7}{192} \operatorname{sen}(6 \arctan(\operatorname{sen}(x) + 2)) + C \end{aligned}$$

$$7. \int \frac{14\cos(x) + 11}{24\cos^2(x) + 38\cos(x) + 5} dx$$

Solución.

Factorizando el denominador

$$I = \int \frac{4\cos(x) + 11}{24 \left(\cos(x) - \frac{-19+\sqrt{241}}{24} \right) \left(\cos(x) - \frac{-19-\sqrt{241}}{24} \right)} dx$$

Para simplificar la expresión haremos que

$$y_1 = \frac{-19 + \sqrt{241}}{24} \quad \wedge \quad y_2 = \frac{-19 - \sqrt{241}}{24}$$

Por fracciones parciales

$$\frac{4\cos(x) + 11}{24(\cos(x) - y_1)(\cos(x) - y_2)} = \frac{A}{\cos(x) - y_1} + \frac{B}{\cos(x) - y_2}$$

$$\frac{4\cos(x) + 11}{24} = A(\cos(x) - y_2) + B(\cos(x) - y_1)$$

Resolviendo el sistema de ecuaciones

$$\begin{cases} A + B = \frac{1}{6} \\ Ay_2 + By_1 = -\frac{11}{24} \end{cases}$$

Entonces

$$A = \frac{4y_1 + 11}{24(y_1 - y_2)} \quad \wedge \quad B = -\frac{4y_2 + 11}{24(y_1 - y_2)}$$

Sustituyendo en la integral

$$\int \frac{4\cos(x) + 11}{24(\cos(x) - y_1)(\cos(x) - y_2)} dx = \frac{4y_1 + 11}{24(y_1 - y_2)} \int \frac{dx}{\cos(x) - y_1} - \frac{4y_2 + 11}{24(y_1 - y_2)} \int \frac{dx}{\cos(x) - y_2}$$

Sean I_1 y I_2 donde

$$I_1 = \int \frac{dx}{\cos(x) - y_1} \quad \wedge \quad I_2 = \int \frac{dx}{\cos(x) - y_2}$$

Se resuelve I_1 ya que I_2 es la misma expresión, sea

$$t = \tan\left(\frac{x}{2}\right) \rightarrow dx = \frac{2}{1+t^2} dt$$

Sustituyendo en I_1

$$I_1 = \int \frac{\frac{2}{1+t^2} dt}{\frac{1-t^2}{1+t^2} - y_1} \rightarrow I_1 = 2 \int \frac{dt}{(1-y_1) - (1+y_1)t^2}$$

$$I_1 = \frac{2}{1-y_1} \int \frac{dt}{1 - \frac{1+y_1}{1-y_1} t^2} \rightarrow I_1 = \frac{2}{1-y_1} \int \frac{dt}{1 - \left(\sqrt{\frac{1+y_1}{1-y_1}} t \right)^2}$$

Recordar que

$$\int \frac{du}{1-u^2} = \operatorname{arctanh}(u) + C$$

Sustituyendo en I_1

$$I_1 = \frac{2}{\sqrt{1-y_1^2}} \operatorname{arctanh} \left(t \sqrt{\frac{1+y_1}{1-y_1}} \right) + C_1$$

Devolviendo a su variable original de igual manera para I_2

$$I_1 = \frac{2}{\sqrt{1-y_1^2}} \operatorname{arctanh} \left(\tan \left(\frac{x}{2} \right) \sqrt{\frac{1+y_1}{1-y_1}} \right) + C_1$$

$$I_2 = \frac{2}{\sqrt{1-y_2^2}} \operatorname{arctanh} \left(\tan \left(\frac{x}{2} \right) \sqrt{\frac{1+y_2}{1-y_2}} \right) + C_2$$

Sustituyendo en la integral I

$$I = \frac{4y_1 + 11}{24(y_1 - y_2)} \left[\frac{2}{\sqrt{1-y_1^2}} \operatorname{arctanh} \left(\tan \left(\frac{x}{2} \right) \sqrt{\frac{1+y_1}{1-y_1}} \right) \right] \\ - \frac{4y_2 + 11}{24(y_1 - y_2)} \left[\frac{2}{\sqrt{1-y_2^2}} \operatorname{arctanh} \left(\tan \left(\frac{x}{2} \right) \sqrt{\frac{1+y_2}{1-y_2}} \right) \right] + C$$

Teniendo en cuenta que

$$y_1 = \frac{-19 + \sqrt{241}}{24} \quad \wedge \quad y_2 = \frac{-19 - \sqrt{241}}{24}$$